Astrea Academy Trust
INSPIRING BEYOND MEASURE

Name .....

Class .....

Tutor Group .....

YEAR 11 MATHEMATICS

Unit 1: Further Algebra PART 2

PART 2: Algebraic Fractions, Functions, Iteration, Proof

$$\frac{x+5}{2x+10}$$

$$x_{n+1} = \sqrt{x_n + 2}$$

$$f(x) = 2x + 3$$

$$f^{-1}(x) = \frac{x-3}{2}$$

KEY VOCABULARY LIST	
Fraction	Fractions represent equal parts of a whole or a collection.
Numerator	The top number in a fraction, represents the parts.
Denominator	The bottom number in a fraction, represents the whole.
Function	A special relationship where each input has a single output, usually written as $f(x)$.
Inverse Function	An inverse function is a function that undoes the action of the original function.
Composite Function	Composite functions are where more than one function is applied to an input, in a specific order.
Formula	A formula is special equation that links two or more variables.
Iteration	Iteration means repeatedly carrying out a process. In this unit, start with an initial value and substitute this into the iteration formula to obtain a new value, then use the new value for the next substitution, and so on.

KEY VOCABULARY LIST	
Solve	To solve means to find a solution. Simultaneous equations have **two** solutions.
Quadratic simultaneous equations	Quadratic simultaneous equations are two or more equations that share variables which include x^2 and y^2
Constant	In algebra, a constant is a number on its own, or sometimes a letter such as a, b or c that stands for a fixed number.
Variable	Variable, in algebra, a symbol (usually a letter) standing in for an unknown numerical value in an equation.
Linear Function	- those whose graph is a straight line. - has the following form. y = mx + c. - has one independent variable and one dependent variable. The independent variable is x and the dependent variable is y.
Simultaneous Equations	Two or more equations that share variables. Example: two equations that share the variables x and y: x + y = 6 −3x + y = 2
Elimination Method	Eliminate" means to remove: this method works by removing variables until there is just one left. The idea is that we **can safely**: **multiply** an equation by a constant (except zero), **add** (or subtract) an equation on to another equation
Substitution Method	These are the steps: Write one of the equations so it is in the style "variable = ..." Replace (i.e. substitute) that variable in the other equation(s). Solve the other equation(s)

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BOOKLET STRUCTURE

- **Start of subtopic:**
  - Read and annotate the information in the Frayer model or introduction text
  - I do (teacher led example).
  - We do (teacher and scholar led example).
  - You do (independent scholar example).
  - Whiteboard check.
  - Independent practice and mixed practice.

- **End of subtopic:**
  - GCSE questions – to ensure you are confident with the style you will encounter in your exams

LEARNING OUTCOME: To simplify and solve algebraic fractions.

Objectives:

1. To be able to simplify basic algebraic fractions.
2. To be able to simplify algebraic fractions by factorising.
3. To multiply and divide with algebraic fractions.
4. To solve equations involving algebraic fractions.

Keywords:
- Fraction
- Solve
- Factorise
- Denominator

An algebraic fraction is a fraction that has an algebraic term/expression in either the numerator, denominator or both. In a previous unit, we learned how to deal with simple algebraic fractions involving only single terms in the numerator or denominator. In this unit we will learn how to solve problems involving more complex algebraic fractions.

There are two types of expressions where it is unclear whether we should classify them as algebraic fractions or not:

- A fraction of the form $\frac{ax}{b}$ can be written as $\frac{a}{b}x$. The first form fits our definition of an algebraic fraction, as it has a term in the denominator. The second form does not: it is just a variable with a fractional coefficient. Whether we will consider these terms as algebraic fractions or not will depend on the context.

- An algebraic fraction that can be simplified. For example:

$$\frac{6x^3y^2}{2xy^2} \equiv 3x^2$$

The fraction on the left has algebraic terms in both its numerator and denominator, so we could consider it an algebraic fraction. The simplified expression is not an algebraic fraction.

In higher tier GCSE, there are two broad categories of 'algebraic fractions' questions: the first involved evaluating and simplifying by accurately applying operations to fractions. The second involves solving equations involving algebraic fractions. Each of these families of questions can be relatively straightforward or extremely complicated – the following pages cover the full range of difficulty.

Fractions

Definition

A fraction represents a proportion - the number of parts of a whole when the whole has been split into equal parts.

Characteristics

- The numerator (top number) represents number of parts.
- The denominator (bottom number) represents how many parts in the whole.
- Each part must be equal.

Examples

= ⅔

= 1

Non - Examples

= ⅕

= 3

Writing in the correct form

Later in this unit we will be asked to express algebraic fractions in a particular form - "write in the form…" is the most common way these questions are asked in your GCSE exam.

The skills required for this involve expanding brackets, factorising, and simplifying. We can practice these skills on individual expressions first, before applying them to algebraic fractions.

Look carefully at the questions in the following task, before deciding on your answer:

Task: Decide if the following expressions are in the correct form. If they are in correct form what are a, b and c?

a) Is $7x + 9$ in the form $ax + b$?
b) Is $5x - 3$ in the form $ax + b$?
c) Is $5(2x + 3)$ in the form $ax + b$?
d) Is $4x^2 - 5x$ in the form $x(ax + b)$?
e) Is $4x^2 - 5x$ in the form $ax^2 + bx$?
f) Is $\frac{3}{5}x + 2$ in the form $ax + b$?
g) Is $x - 3$ in the form $ax + b$?
h) If $4(x + 4) + 5(x + 1)$ in the form $a(bx + c)$?

	Example 1 - I do…	*Example 2 - We do…*	*Example 3 - You do…*
Steps	Express $6(2x - 3)$ in the form $ax + b$	Express: $3(x + 2) + 6(3x + 5)$ in the form $a(bx + c)$	Express $12x^2 - 18x^5$ in the form $ax^2(b + cx^3)$
1) Expand and collect like terms			
2) Check the required form and you may need to factorise			

INDEPENDENT PRACTICE

Question 1: Express in the form $ax + b$

a) $7(2x + 3)$	**b)** $-2(x + 4)$	**c)** $\frac{1}{2}(6x + 8)$	**d)** $-8(4 - 3x)$

Question 2: Express in the form $ax(bx + c)$

a) $45x^2 + 18x$	**b)** $-6x^2 + 12x$	**c)** $32x - 28x^2$	**d)** $5x(x + 3) + 6x$

Simplifying basic algebraic fractions

The rules for simplifying algebraic fractions are like those for simplifying a normal fraction: divide (or occasionally multiply) the numerator and denominator by the same amount.

We begin below by simplifying fractions with only single terms in numerator and denominator:

	Example 1 - I do...	*Example 2 - We do...*	*Example 3 - You do...*
Steps	Simplify $\frac{12x^4y^3}{9xy^2}$	Simplify $\frac{28x^5y}{49x^5y^3}$	Simplify $\frac{30x^5y^3}{18x^{10}y^3}$
1) Check if the numerator/denominator for common factors.			
2) Cancel common factors, being careful if powers are involved			

INDEPENDENT PRACTICE

1. Simplify these algebraic fractions:

a) $\frac{3}{12}$

b) $\frac{3x}{12}$

c) $\frac{3}{12x}$

d) $\frac{12x}{6}$

e) $\frac{12x}{6x}$

f) $\frac{12x^2}{3}$

g) $\frac{12x^2}{3x^4}$

h) $\frac{12a^3x^2}{3x^2}$

i) $\frac{12x}{3x^2}$

j) $\frac{10x^7y}{12x}$

2. Explain the mistake in the following simplification:

a)

$$\frac{7x^8}{10x^2} \rightarrow \frac{7x^4}{10}$$

_____

_____

Simplifying simple algebraic fractions using factorisation

Simplifying Algebraic Fractions will often use your knowledge of factorising – either into singe brackets or double brackets.

Knowledge Check

Factorise these expressions:

a) $4x + 12$

b) $3x - 9$

c) $6x - 8$

d) $12x + 30$

e) $x^2 + 4x + 3$

f) $y^2 + 9y + 8$

g) $n^2 + 6n + 5$

h) $k^2 + 12k + 35$

i) $x^2 - 4$

j) $y^2 - 16$

k) $x^2 - 49$

l) $x^2 - 121$

m) $2x^2 - 3x - 9$

n) $3x^2 + 8x + 4$

o) $4x^2 - 9x + 2$

p) $3x^2 + 10x + 7$

Steps	Example 1 - I do… Simplify $\frac{2(x+2)}{x+2}$	Example 2 - We do… Simplify $\frac{(x+3)}{2(x+3)}$	Example 3 - You do… Simplify $\frac{4(x-3)}{(x-3)(x+2)}$
1) Check if the numerator/denominator need to be factorised.			
2) Factorise.			
3) Cancel common factors.			

INDEPENDENT PRACTICE

1. Simplify the algebraic fractions.

a) $\frac{2(x+3)}{3(x+3)}$

b) $\frac{2(x+1)}{(x+3)}$

c) $\frac{(x+3)}{2(x-3)}$

d) $\frac{x+3}{2(x+3)}$

e) $\frac{(x+3)(x+6)}{(x+3)}$

f) $\frac{(x-1)(x+1)}{(x-1)}$

g) $\frac{(x-3)}{(x-4)(x-3)}$

h) $\frac{(x-3)(x+2)}{(x+2)(x+9)}$

i) $\frac{(x+3)(x+6)(x+2)}{(x+3)(x+2)(x-4)}$

j) $\frac{(x-3)(2x-1)(x+2)}{(x-3)(x+2)(2x-1)}$

Steps	Example 4 - I do… Simplify $\frac{5x+10}{x+2}$	Example 5 - We do… Simplify $\frac{5x+15}{2(x+3)}$	Example 6 - You do… Simplify $\frac{4x+12}{2x+6}$
1) Check if the numerator/denominator need to be factorised.			
2) Factorise.			
3) Cancel common factors.			

INDEPENDENT PRACTICE 2. Simplify the algebraic fractions.

a) $\frac{2x+2}{3(x+1)}$

b) $\frac{2x+2}{3x+3}$

c) $\frac{3x+3}{2(x+1)}$

d) $\frac{x+3}{2(x+3)}$

e) $\frac{4x+12}{3(x+3)}$

f) $\frac{4x+12}{(x+1)(x+3)}$

g) $\frac{x+3}{4x+12}$

h) $\frac{5x+15}{x+3}$

Steps	Example 7 - I do… Simplify $\frac{x^2+5x+6}{(x+3)(x+4)}$	Example 8 - We do… Simplify $\frac{x^2-x-12}{2x+6}$	Example 9 - You do… Simplify $\frac{x^2-x-12}{2x-8}$
1) Check if the numerator/ denominator need to be factorised.			
2) Factorise.			
3) Cancel common factors.			

i) $\frac{(x+3)(x+6)}{2x+6}$	j) $\frac{(x+3)(x+4)}{2x+6}$	k) $\frac{(x+3)(x+4)}{2x+8}$	l) $\frac{x^2+7x+12}{2x+8}$
m) $\frac{x^2+7x+12}{(x+4)(x-9)}$	n) $\frac{x^2+7x+12}{x^2-5x-36}$	o) $\frac{2x^2+14x+24}{x^2-5x-36}$	p) $\frac{2x^2+14x+24}{3x^2-15x-108}$
q) $\frac{2x^2+14x+24}{3x^2+4x-15}$	r) $\frac{14x-24-2x^2}{3x^2+4x-15}$	s) $\frac{14x-24-2x^2}{3x^2-4x-15}$	t) $\frac{10x^2-23x+12}{4x^2+4x-15}$

Multiplying and Dividing Algebraic Fractions

Multiplying and diving single term algebraic fractions works similarly to the numerical case:

- For multiplication, the new numerator is the product of the numerators, the new denominator is the product of the denominators. Once the numerator and denominator are calculated, simplify if possible.
- For division we need to rewrite the calculation so that we are multiplying by the reciprocal of the second fraction.

Knowledge Check: Cross Simplification

a) $\frac{3}{15} \times \frac{5}{11} =$

b) $\frac{5}{14} \times \frac{21}{15} =$

c) $\frac{13}{80} \times \frac{20}{39} =$

	Example 13 - I do...	***Example 14 - We do...***	***Example 15 - You do...***
Steps	Simplify $\frac{2b}{3} \times \frac{4}{b}$	Simplify $\frac{b^2}{2} \times \frac{8}{b}$	Simplify $\frac{2b^3}{3} \times \frac{4}{2b}$
1) Cross cancel.			
2) Simplify			

INDEPENDENT PRACTICE 3. Simplify the algebraic fractions.

a) $\frac{b}{4} \times \frac{4}{b}$

b) $\frac{b^2}{4} \times \frac{4}{b}$

c) $\frac{b^3}{4} \times \frac{4}{2b^2}$

d) $\frac{b^3}{4} \times \frac{8}{2b^2}$

e) $\frac{8}{b^3} \times \frac{2b^2}{4}$

Steps	Example 16 - I do… Simplify $\frac{2(b+2)}{2c} \times \frac{4c}{b+2}$	Example 17 - We do… Simplify $\frac{2(a+4)}{2c} \times \frac{4c}{2(a+4)}$	Example 18 - You do… Simplify $\frac{3(c+2)}{8c} \times \frac{2c(c-1)}{c+2}$
1) Cross cancel.			
2) Simplify			

INDEPENDENT PRACTICE

4. Simplify the algebraic fractions.

a) $\frac{2(x+2)}{2c} \times \frac{4c}{x+2}$

b) $\frac{2(x-2)}{2c} \times \frac{4c}{x-2}$

c) $\frac{2(x+2)}{2c^3} \times \frac{4c^2}{x+2}$

d) $\frac{2x+4}{2c} \times \frac{4c^3}{x+2}$

Steps	Example 16 - I do… $\frac{x^2+5x+6}{4} \times \frac{2}{x+2}$	Example 17 - We do… $\frac{x^2+5x+6}{x^2+9x+20} \times \frac{x+4}{x+2}$	Example 18 - You do… $\frac{x^2+x-6}{x^2+9x+20} \times \frac{x+4}{x-2}$
1) Where possible, factorise any parts of the expression			
2) Look to see if any terms can be cancelled out.			
3) Simplify as appropriate.			

INDEPENDENT PRACTICE 5. Simplify the algebraic fractions.

a) $\frac{x^2+3x-4}{x+2} \times \frac{x+2}{6(x+4)}$

b) $\frac{x^2+3x-4}{x+2} \times \frac{x+2}{6x+24}$

c) $\frac{x+2}{x^2+3x-4} \times \frac{6x+24}{x+2}$

d) $\frac{(x+1)^2}{x^2+3x-4} \times \frac{x-1}{x+1}$

e) $\frac{x^2+16x+63}{x+3} \times \frac{x+3}{(x+9)(x-14)}$

f) $\frac{x^2+16x+63}{x+3} \times \frac{x+3}{x^2+9x}$

g) $\frac{x^2+16x+63}{3x+9} \times \frac{x+3}{x^2+9x}$

h) $\frac{3x+9}{x^2+16x+63} \times \frac{x^2+9x}{x+3}$

i) $\frac{x^2-9}{x+4} \times \frac{2x^2+8x}{x+3}$

j) $\frac{x^2-36}{x+6} \times \frac{2x}{x-6}$

k) $\frac{x^2+x-6}{x^2-25} \times \frac{x^2+10x+25}{x^2-4}$

l) $\frac{3x^2+8x-3}{25} \times \frac{30}{6x^2+13x-5}$

Steps	Example 19 - I do… Simplify $\frac{b}{3} \div \frac{6}{7b}$	Example 20 - We do… Simplify $\frac{x-4}{8} \div \frac{3(x-4)}{2}$	Example 21 - You do… Simplify $\frac{x^2+4x+4}{8} \div \frac{2(x+2)}{2}$
1) Multiply the first fraction by the **reciprocal** of the second.			
2) Check if it can be factorised.			
3) Cross Cancel and simplify			

INDEPENDENT PRACTICE

6. Simplify the algebraic fractions.

a. $\frac{2x}{7} \div \frac{4y}{14}$

b. $\frac{4y^2}{9x} \div \frac{2y}{3x^2}$

c. $\frac{x-3}{15} \div \frac{5}{2x-6}$

d. $\frac{2x+1}{2} \div \frac{4x+2}{4}$

e. $\frac{x}{6} \div \frac{2x^2+x}{3}$

f. $\frac{x-2}{12} \div \frac{4}{x-3}$

g. $\frac{x-5}{10} \div \frac{x^2-5x}{5}$

h. $\frac{3x+9}{2} \div \frac{x+3}{4}$

INDEPENDENT PRACTICE

i. $\frac{4}{x-2} \div \frac{3}{x^2-2x}$

j. $\frac{11}{4x^2+2x} \div \frac{3}{2x+1}$

k. $\frac{x+3}{x+1} \div \frac{x}{(x+1)^2}$

l. $\frac{x^2}{7} \div \frac{6x^3+8x^2}{x^2-7x}$

m. $\frac{x^2+7x+10}{2} \div \frac{x^2+4x-5}{4}$

n. $\frac{14}{x^2-5x+6} \div \frac{7}{x^2+3x-10}$

o. $\frac{x+4}{x+5} \div \frac{3x+12}{x^2-25}$

p. $\frac{x^2+4x+4}{x} \div \frac{(x+2)}{4x(x-2)}$

q. $\frac{x^2-2x-15}{x+2} \div \frac{x^2-11x+30}{2x}$

r) $\frac{2y^2+4y}{3y^2+7y+2} \div \frac{9y^2-1}{3y^2-y} =$

s) $\frac{x^2+4x-12}{x^2-25} \div \frac{x+6}{x^2-5x} =$

t) $\frac{3x^2-5x-2}{x^2-4x} \div \frac{3x+1}{x^2-8x+16}$

u) $\frac{3x^2+8x-3}{25} \div \frac{6x^2+13x-5}{30}$

v) $\frac{x^3-x}{x^4-1} \div \frac{x+1}{x^3+x}$

Adding and Subtracting Algebraic Fractions

When the algebraic fractions we want to add or subtract have different denominators, it is necessary to find a common denominator. As with numerical fractions we do this by finding the lowest common multiple of the denominators.

Knowledge Check

Evaluate and simplify the following:

a) $\frac{1}{4}+\frac{1}{3}$

b) $\frac{1}{6}+\frac{2}{5}$

c) $\frac{3}{10}-\frac{1}{5}$

d) $\frac{3}{7}+\frac{2}{5}$

e) $\frac{5}{6}-\frac{5}{7}$

	Example 22 - I do...	***Example 23 - We do...***	***Example 24 - You do...***
Steps	Simplify $\frac{x}{5}+\frac{3x}{8}$	Simplify $\frac{x}{7}+\frac{3x}{8}$	Simplify $\frac{2x}{5}+\frac{3x}{7}$
1) Find the Lowest common denominator			
2) Multiply the numerators.			
3) Write in simplest form.			

INDEPENDENT PRACTICE

a) $\frac{3x}{2}+\frac{4x}{5}$

b) $\frac{x}{2}+\frac{4x}{5}$

c) $\frac{x}{2}+\frac{x}{5}$

d) $\frac{x}{2}-\frac{x}{5}$

e) $\frac{5x}{5}-\frac{x}{2}$

	Example 25 - I do...	Example 26 - We do...	Example 27 - You do...
Steps	Simplify $\frac{x+1}{2}+\frac{x+3}{3}$	$\frac{5}{x+7}+\frac{1}{2x-1}$	Simplify $\frac{x+5}{3}+\frac{x-2}{4}$
1) Find the Lowest common denominator			
2) Multiply the numerators and expand and simplify.			
3) Write in simplest form.			

INDEPENDENT PRACTICE 7. Simplify the algebraic fractions.

a) $\frac{x+2}{4}+\frac{x+2}{5}$

c) $\frac{x-3}{4}+\frac{x+2}{3}$

d) $\frac{3}{x-1}+\frac{4}{3x+2}$

e) $\frac{x-1}{5}+\frac{x+2}{3}$

f) $\frac{x-4}{4}+\frac{2x-3}{2}$

g) $\frac{3x+2}{4}-\frac{x-2}{5}$

h) $\frac{4x+2}{2}-\frac{x+3}{3}$

i) $\frac{2x+1}{4}-\frac{2x-1}{3}$

j) $\frac{2x+1}{2} - \frac{2x+1}{4}$

k) $\frac{3x-2}{2} - \frac{4x-3}{4}$

l) $\frac{5x-4}{4} - \frac{2x-3}{2}$

m) $\frac{7x+5}{5} - \frac{2x+1}{3}$

Example 28 - I do... ***Example 29 - We do...*** ***Example 30 - You do...***

Steps	Simplify $\frac{5}{x} + \frac{8}{3x}$	Simplify $\frac{7}{2x} + \frac{2}{x^2}$	Simplify $\frac{7}{3x} + \frac{5}{x^2}$
1) Find the Lowest common denominator			
2) Multiply the numerators and expand and simplify.			
3) Write in simplest form.			

INDEPENDENT PRACTICE

8. Simplify the algebraic fractions.

a) $\frac{5}{x} + \frac{2}{x}$

b) $\frac{5}{x^2} + \frac{2}{x}$

c) $\frac{5}{2x^2} - \frac{2}{x}$

d) $\frac{6}{2x^2} - \frac{2}{x}$

When we are adding and subtracting algebraic fractions, it is sometimes necessary to either expand double brackets or factorise quadratics.

Steps	***Example 31 - I do…***	***Example 32 - You do…***
	Express the following as a single simplified fraction. $\frac{x-3}{x+2}+\frac{x+5}{x-2}$	Express the following as a single simplified fraction. $\frac{x+7}{x-3}+\frac{x-2}{x+4}$
1) Find the Lowest common denominator		
2) Multiply the numerators and expand and collect terms..		
3) Check if any final simplification is possible .		

Steps	***Example 33 - I do…***	***Example 34 - You do…***
	Express the following a single simplified fraction. $\frac{10x+9}{x^2+6x+5}-\frac{3}{x+5}$	Express the following a single simplified fraction. $\frac{4x+1}{x^2-2x-15}-\frac{3}{x+3}$
1) Find the Lowest common denominator		
2) Multiply the numerators and expand and collect terms..		
3) Check if any final simplification is possible .		

Question 9. Write as a single fraction, simplify where possible.

a)	$\frac{x-7}{2x-1}+\frac{3x}{x-2}$	**b)**	$\frac{x+1}{x-2}-\frac{3x+4}{2}$
c)	$\frac{4}{x^2+2x+1}+\frac{2}{x+1}$	**d)**	$\frac{x^2+1}{x^2-x-6}-\frac{2x}{x-3}$
e)	$\frac{8}{x^2+4x+2}+\frac{5}{x^2-4}$	**f)**	$\frac{5x}{x^2+5x+6}-\frac{7}{x^2+x-6}$

Q1.

Show $\dfrac{x^2 - x - 6}{2x^2 - 5x - 3}$ that can be written in the form $\dfrac{ax + b}{cx + d}$ where *a*, *b*, *c* and *d* are integers.

(Total for question = 3 marks)

Q2.

Show that $\dfrac{4x + 3}{2x} + \dfrac{3}{5}$ can be written in the form $\dfrac{ax + b}{cx}$ where *a*, *b* and *c* are integers.

(Total for question = 3 marks)

Q3.

(a) Write $\dfrac{5}{x + 1} + \dfrac{2}{3x}$ as a single fraction in its simplest form.

.....

(2)

Q4.

(a) Express $\frac{x}{x+2} + \frac{2x}{x-4}$ as a single fraction in its simplest form.

.....

(3)

Q6.

$\frac{2x+3}{x-5} + \frac{x-4}{x+5} - 3$ can be written in the form $\frac{ax+b}{x^2-25}$ where *a* and *b* are integers.

Work out the value of *a* and the value of *b*.
You must show all your working.

a = ...

b = ...

(Total for question = 3 marks)

Q7.

(a) Simplify $\dfrac{x^2 - 16}{2x^2 - 5x - 12}$

.....

(3)

Q8.

(a) Write $\dfrac{4x^2 - 9}{6x + 9} \times \dfrac{2x}{x^2 - 3x}$ in the form $\dfrac{ax + b}{cx + d}$ where *a*, *b*, *c* and *d* are integers.

.....

(3)

(b) Express $\dfrac{3}{x+1} + \dfrac{1}{x-2} - \dfrac{4}{x}$ as a single fraction in its simplest form.

.....

(3)

(Total for question = 6 marks)

Q9.

Show that $\dfrac{7x-14}{x^2+4x-12} \div \dfrac{x-6}{x^3-36x}$ simplifies to *ax* where *a* is an integer.

(Total for question = 4 marks)

Q10.

Show that $\dfrac{6x-y}{10xy} + \dfrac{1}{2x} - \dfrac{2y-7x}{5xy}$ simplifies to $\dfrac{k}{y}$ where *k* is an integer.

(Total for question = 3 marks)

Subtopic 4b – Algebraic Fractions: Solving

We can solve equations involving algebraic fractions in one of two ways:

1. Simplify the equation so that each side is either a single algebraic fraction or an expression. Multiply through by the denominators of any algebraic fractions – leaving an equation with no algebraic fractions to be solved.
2. Calculate the lowest common multiple of the denominators of the algebraic fractions. Multiply through by this expression to again reach an equation free of algebraic fractions.

Before we do this, we need to recap the solving of equations involving fractions that you have seen before. Attempt the questions below independently, showing clearly what you do for the steps at each stage.

Knowledge Check

a) $\frac{x+5}{7} = 5$

b) $\frac{3x-2}{5} = 4$

c) $\frac{x-4}{7} = 3$

d) $3 = \frac{2y+1}{5}$

e) $\frac{5a+3}{4} = 7$

f) Solve $\frac{3x}{5} = \frac{x+1}{10}$

g) Solve $\frac{x+4}{2} = \frac{x+10}{3}$

h) Solve $\frac{x-2}{3} = \frac{x+4}{5}$

Steps	Example 31 - I do… Solve $\frac{5y+8}{6} - \frac{3y+2}{4} = 1$	Example 32 - We do… Solve $\frac{6x-1}{4} - \frac{5-2x}{2} = 7$
1) Multiply through to make the denominators the same. (or: clear the fraction by multiplying both sides appropriately)		
2) Expand and simplify the numerators.		
3) Solve the equation.		

Steps	Example 33 - You do… Solve $\frac{x-1}{2} + \frac{2x+3}{4} = 1$	Example 34 - You do… Solve $\frac{2x+1}{3} + \frac{x-5}{2} = 4$
1) Multiply through to make the denominators the same. (or: clear the fraction by multiplying both sides appropriately)		
2) Expand and simplify the numerators.		
3) Solve the equation.		

INDEPENDENT PRACTICE

1. Solve the algebraic fractions.

a) $\frac{x}{3}+\frac{x-2}{6}=4$	b) Solve $\frac{x+8}{2}+\frac{x+6}{4}=7$
c) $\frac{x+1}{2}+\frac{x+2}{5}=3$	d) $\frac{x+1}{2}+\frac{2x+1}{3}=9$

e) $\frac{x+1}{2}+\frac{3x-1}{4}=4$	f) $\frac{4x+1}{3}-\frac{x+2}{4}=2$
g) $\frac{2x+1}{2}-\frac{x+1}{7}=1$	h) $\frac{3x+1}{5}-\frac{5x-1}{7}=0$

Steps	Example 35 - I do… Solve $\frac{4}{x+6}+\frac{5}{x+8}=1$	Example 36 - We do… Solve $\frac{3}{x-6}+\frac{4}{x-9}=1$
1) Multiply through the make the denominators the same.		
2) Expand and simplify the numerators.		
3) Multiply through by the denominator.		
4) Expand and simplify the RHS.		
5) Rearrange into form $ax^2+bx+c=0$		
6) Solve the quadratic equation.		

Steps	Example 37 - You do… Solve $\frac{2}{x+6}+\frac{4}{x+8}=2$	Example 38 - You do… Solve $\frac{4}{x-5}+\frac{2}{x-8}=3$
1) Multiply through the make the denominators the same.		
2) Expand and simplify the numerators.		
3) Multiply through by the denominator.		
4) Expand and simplify the RHS.		
5) Rearrange into form $ax^2+bx+c=0$		
6) Solve the quadratic equation.		

INDEPENDENT PRACTICE

10. Solve the algebraic fractions.

a) $\frac{2}{x-9} + \frac{4}{x-7} = 2$

$x =$ _____ or $x =$ _____

b) $\frac{3}{x-1} + \frac{2}{x-3} = 1$

$x =$ _____ or $x =$ _____

c) $\frac{1}{2x-7} + \frac{6}{x+4} = 1$

$x =$ _____ or $x =$ _____

d) $\frac{1}{2x-5} + \frac{4}{x+2} = 1$

$x =$ _____ or $x =$ _____

e) $\frac{5}{2x-3} + \frac{2}{x-6} = 1$

$x =$ _____ or $x =$ _____

f) $\frac{2}{5x+2} + \frac{5}{x+4} = 1$

$x =$ _____ or $x =$ _____

INDEPENDENT PRACTICE

10. Solve the algebraic fractions.

g) $\frac{4}{2x-1} + \frac{6}{4x-7} = 2$

$x =$ _____ or $x =$ _____

h) $\frac{5}{2x-6} + \frac{3}{2x+6} = 1$

$x =$ _____ or $x =$ _____

i) $\frac{6}{4x+1} + \frac{4}{2x+3} = 2$

$x =$ _____ or $x =$ _____

j) $\frac{2}{4x-9} + \frac{4}{2x-3} = 2$

$x =$ _____ or $x =$ _____

k) $\frac{6}{2x-3} + \frac{4}{3x-7} = 2$

$x =$ _____ or $x =$ _____

l) $\frac{3}{4x+7} + \frac{6}{2x+5} = 3$

$x =$ _____ or $x =$ _____

1) Show that $\frac{4}{x}+\frac{2}{x-1}$ simplifies to $\frac{6x-4}{x(x-1)}$

b) Hence or otherwise, solve $\frac{4}{x}+\frac{2}{x-1}=3$, giving your solutions correct to 3sf.

$x =$_____or $x =$ _____

2)

a. Prove that the equation $\frac{7}{x+1}+\frac{4}{x+4}=3$ simplifies to $3x^2+4x-20=0$.

b. Hence solve the equation $\frac{7}{x+1}+\frac{4}{x+4}=3$.

$x =$_____or $x =$ _____

3) Emma swam 300 m at a speed of $(x-1)$ m/s, then a further 200 m at $(x-2)$ m/s.

The total swim took Emma 10 minutes.

a. Show that $6x^2-23x+20=0$.

b. Find the speed at which Emma swam the first 200 m.

Q1.

Solve $\frac{3x-2}{4} - \frac{2x+5}{3} = \frac{1-x}{6}$

x = ...

(Total for question = 4 marks)

Q2.

(b) Solve $\frac{2}{x+1} + \frac{x}{2x+3} = 1$

Give your solutions as surds.

.....

(4)

Q3.

Solve $\frac{x+1}{3} + \frac{2x+5}{4} = 2$

x = ...

(Total for question = 4 marks)

Q4.

Solve $\frac{1}{x+4} + \frac{3}{2-2x} = 1$

.....

(Total for question = 4 marks)

LEARNING OUTCOME: To work with functions.

Objectives:

1. To understand what functions are
2. To be able to find inverse functions.
3. To be able to work with composite functions.

Keywords:
- Function
- Inverse
- Composite

Previously when studying mathematics, you have come across many functions, but they have not been in the formal function notation. You have seen them in this format.

$$y = 3x - 5, \qquad y = x^3, \qquad y = 23 - x^2$$

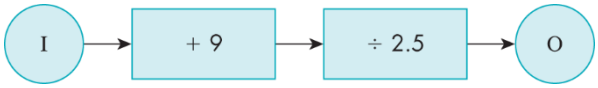
Formal function notation replaces the variable y with the notation $f(x)$ (we say "f of x"), then $g(x)$,$h(x)$, etc…

We refer to these as the functions f, g and h respectively and they could look like the following:

$$f(x) = 3x - 5, \qquad g(x) = x^3, \qquad h(x) = 23 - x^2$$

Prior Knowledge Check

1. Use the function machine to find the output for each input.



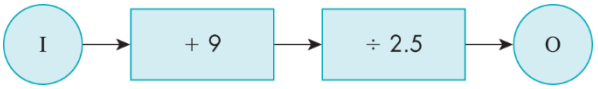
i. 11

ii. 26

iii. – 4

iv. 1

2. Use the function machine to find the input for each output.



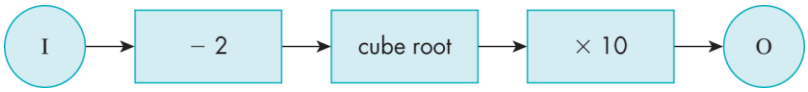
i. 18

ii. 0

iii. 500

iv. 3

3. Use the function machine to find the input for each output.



i. 30

ii. 60

iii. 18

Substituting into Functions

Example 1 - I do…

$f(x) = 2x + 1$

Find:

$f(3)$

Substitute 3 into the function.

$f(3) = 2(3) + 1$

$f(3) =$

Example 2 - We do…

$f(x) = 2x + 1$

Find:

$f(-7)$

Substitute -7 into the function.

$f(-7) =$

Example 3- You do…

$f(x) = x^2 - 2$

Find:

$f(8)$

$f(8) =$

INDEPENDENT PRACTICE

1. f(x) = $2x^2 + 7$ Find the value of each of these

a) $f(3)$

b) $f(11)$

c) $f(-2)$

d) $f(30)$

e) $f(-30)$

f) $f(\sqrt{3})$

2. g(x) = $25 - x^2$ Find the value of each of these

a) $g(4)$

b) $g(-8)$

c) $g(9)$

d) $g(-9)$

e) $g(\sqrt{13})$

f) $g(-0.4)$

3. f(x) = $2x^2 + 3x$ Find the value of each of these

a) $f(-6)$

b) $f(4)$

4. g(x) = $x^2 + 3x - 7$ Find the value of each of these

a) $g(3)$

b) $g(-4)$

FINDING THE INPUT

Steps	Example 4- I do… Given that $f(x) = 6x + 1$ Find the value of x when $f(x) = 7$	Example 5 - We do… Given that $f(x) = 10x + 3$ Find the value of x when $f(x) = 7$	Example 6 - You do… Given that $f(x) = 8x + 5$ Find the value of x when $f(x) = 7$
1) Set the function and the output equal and set up an equation.			
2) Solve for the input.			

INDEPENDENT PRACTICE Given that $f(x) = 6x + 1$, find the value of x when:

$a)\ f(x) = 1$ $b)\ f(x) = 4$ c) $f(x) = 3$ d) $f(x) = 10$

$e)\ f(x) = -3$ $f)\ f(x) = -4$ g) $f(x) = -2$ h) $f(x) = -10$

Example 7- I do…

$f(x) = 2x + 1$

Find:

$2f(x)$

Example 8 - We do…

$f(x) = 3x - 1$

Find:

$3f(x)$

Example 9- You do…

$f(x) = x^2 - 2$

Find:

$3f(x)$

INDEPENDENT PRACTICE

1) $f(x) = x^2 + 6$

Find:

a) $2f(x)$

b) $4f(x)$

2) $f(x) = 3x + 2$

Find:

a) $2f(1)$

b) $3f(2)$

Example 10- I do…

$f(x) = 2x + 1$

Find:

$f(x + 1)$

Example 11 - We do…

$f(x) = 3x - 1$

Find:

$f(x - 2)$

Example 12- You do…

$f(x) = x^2 - 2$

Find:

$f(x + 6)$

INDEPENDENT PRACTICE

1) $f(x) = 5x - 3$

Find:

a) $f(x + 2)$

b) $f(x - 4)$

2) $f(x) = x^2 + 5$

Find:

a) $f(x + 2)$

b) $f(x - 1)$

COMPOSITE FUNCTIONS

A composite function is a combination of two functions where one is substituted into another. The notation for this is as such $fg(x)$ (f of g of x). The order in which we find and use composite functions is not commutative, for example $fg(x) \neq gf(x)$. When working with composite we are working from the inside out as is shown in the following example, $fg(x) = f\big(g(x)\big)$, so we evaluate whatever $g(x)$ is first then substitute this into $f(x)$ as the new input.

	Example 13- I do…	*Example 14 - We do…*	*Example 15 - You do…*
Steps	$f(x) = 2x$ $g(x) = x + 1$ Find $fg(2)$	$f(x) = 3x$ $g(x) = x + 3$ Find $fg(2)$	$f(x) = 3x$ $g(x) = x - 2$ Find $fg(3)$
1) Substitute the number into the first function.			
2) Substitute the output into the second function.			

INDEPENDENT PRACTICE

7. Given that $f(x) = 2x$ and $g(x) = x + 1$ Find:

a) $fg(3)$	b) $fg(4)$	c) $gf(2)$	d) $gf(3)$
e) $fg(3)$	f) $ff(2)$	g) $gg(2)$	h) $ggg(3)$

8. Given that $f(x) = 2x$ and $g(x) = x - 1$ Find:

a) $fg(2)$	b) $fg(4)$	c) $gf(2)$	d) $gf(5)$

	Example 16- I do...	***Example 17 - We do...***	***Example 18 - You do...***
Steps	$f(x) = 2x$ $g(x) = x + 1$ Find $fg(x)$	$f(x) = 3x$ $g(x) = x + 3$ Find $gf(x)$	$f(x) = 3x$ $g(x) = x - 2$ Find $fg(x)$
1) Substitute one function into the other.			
2) Simplify where possible.			

INDEPENDENT PRACTICE

Find the value of these composite functions algebraically.

9) $f(x) = 2x$
$g(x) = x + 1$

a) $fg(x)$	b) $gf(x)$	c) $ff(x)$	d) $gg(x)$

INDEPENDENT PRACTICE

Find the value of these composite functions algebraically.

10) $f(x) = 3x$
$g(x) = x + 1$

a) $fg(x)$

b) $gf(x)$

c) $ff(x)$

d) $gg(x)$

11) $f(x) = 5x$
$g(x) = 3x - 2$

e) $fg(x)$

f) $gf(x)$

g) $ff(x)$

h) $gg(x)$

12) $f(x) = x^2$
$g(x) = 3x - 3$

i) $fg(x)$

j) $gf(x)$

k) $ff(x)$

l) $gg(x)$

13) $f(x) = x^3 - 6$
$g(x) = 4(x - 2)$
$h(x) = 3 - x$

m) $gf(x)$	n) $hg(x)$	o) $fh(x)$	p) $gg(x)$

14) $f(x) = 2x + 1$
$g(x) = \frac{1}{x}$

q) $fg(x)$	r) $gf(x)$	s) $ff(x)$	t) $gg(x)$

15) Let $f(x) = x + 1$ and $g(x) = x^2 + 1$.
If $gf(x) = 17$, determine the possible values of x

INVERSE FUNCTIONS

An inverse function performs the inverse operations of the function that was initially given. It is essentially the same as when we placed outputs into our functions machine at the beginning of this topic. The formal notation for the inverse function is $f^{-1}(x)$, where the -1 does not act as an index, it simply refers to the inverse of the function.

Note if you found the composite function of $ff^{-1}(x)$ or $f^{-1}f(x)$the answer will always be x. This is because you are finding the function and then its inverse (or vice versa), so undoing the function you have just evaluated and ending up where you started.

To find the inverse of a function,$f(x)$, we do this in three steps:

- Replace $f(x)$ with y, so you have an equation
- Then rearrange the equation to make x the subject,
- Then rewrite the equation as $f^{-1}(x) =$ and swap the xs & ys.

	Example 19- I do…	***Example 20 - We do…***	***Example 21- You do…***
Steps	$f(x) = 2x$ Find $f^{-1}(x)$	$f(x) = \frac{x-3}{2-x}$, find $f^{-1}(x)$	$f(x) = 2x - 2$ Find $f^{-1}(x)$
1) Write as $f(x)$ as y			
2) Change to make x the subject.			
3) Swap the variables to write as $f^{-1}(x)$			

INDEPENDENT PRACTICE

6. Find the inverse functions $f^{-1}(x)$ of the following functions

a) $f(x) = 4x$	b) $f(x) = 5x$	c) $f(x) = x + 7$	d) $f(x) = 7x + 2$

i) $f(x) = x + 5$

j) $f(x) = x - 1$

k) $f(x) = x - 2$

l) $f(x) = x - 3$

m) $f(x) = 4x - 5$

n) $f(x) = x^3 + 2$

o) $f(x) = \frac{10}{x+1}$

p) $f(x) = 10 - 2x$

q) $f(x) = \frac{x-7}{6}$

r) $f(x) = \frac{3}{x} + 5$

s) $f(x) = \frac{x+2}{3x-5}$

t) $f(x) = \frac{3x+8}{4x-3}$

u) Prove that if $f(x) = \frac{ax+b}{cx-a}$ then $f^{-1}(x) = \frac{ax+b}{cx-a}$

SOLVING QUADRATIC EQUATIONS WITH FUNCTIONS

	Example 22 - I do...	*Example 23 - We do...*	*Example 24- You do...*
Steps	Given that $f(x) = x^2 + 4x$ Solve $f(x) = -3$	Given that $f(x) = x^2 + 4x$ Solve $f(x) = x$	Given that $f(x) = x^2 + 7x$ Solve $f(x) = -12$
1) Set the function and the output equal to each other.			
2)Rearrange into the form $ax^2 + bx + c = 0$			
3) Solve the quadratic equation.			

INDEPENDENT PRACTICE

16) Solve each quadratic equation from the function.

a) $f(x) = x^2 + 4x$

Solve $f(x) = 12$

$x =$ _____ or $x =$ _____

b) $f(x) = x^2 - 12$

Solve $f(x) = x$

$x =$ _____ or $x =$ _____

c) $f(x) = 7x - 10$

Solve $f(x) = x^2$

$x =$ _____ or $x =$ _____

d) $f(x) = x^2 - 6$

Solve $f^{-1}(x) = x$

$x =$ _____ or $x =$ _____

e) $f(x) = \sqrt{x+3}$

Solve $f^{-1}(x) = 6$

$x =$ _____ or $x =$ _____

f) $f(x) = 2x - 15$

Solve $f^{-1}(x) = x^2$

$x =$ _____ or $x =$ _____

g) $f(x) = x^2 - 6$
$g(x) = 3x + 4$

Solve $f(x) = g(x)$

$x =$ _____ or $x =$ _____

h) $f(x) = x^2 - 4$
$g(x) = 4x - 1$

Solve $f(x) = g(x) + 2$

$x =$ _____ or $x =$ _____

i) $f(x) = \sqrt{x+4}$
$g(x) = x - 8$

Solve $f^{-1}(x) = g^{-1}(x)$

$x =$ _____ or $x =$ _____

Q1.

f and *g* are functions such that

$$f(x) = \frac{2}{x^2} \qquad \text{and} \qquad g(x) = 4x^3$$

(a) Find *f*(–5)

.....

(1)

(b) Find fg(1)

.....

(2)

(Total for question = 3 marks)

Q2.

f(x) = 4sin$x$°

(a) Find f(23)

Give your answer correct to 3 significant figures.

.....

(1)

g(x) = 2x – 3

(b) Find fg(34)

Give your answer correct to 3 significant figures.

.....

(2)

h(x) = (x + 4)^2

Ivan needs to solve the following equation h(x) = 25

He writes

(x + 4)^2 = 25
x + 4 = 5
x = 1

This is not fully correct.

(c) Explain why.

.....

.....

(1)

(Total for question = 4 marks)

Q3.

The functions f and g are given by

$$f(x) = \frac{12}{x+1} \quad \text{and} \quad g(x) = 5 - 3x$$

(a) Find f(–3)

.....

(1)

(b) Find fg(1)

.....

(2)

(c) Find $g^{-1}(4)$

.....

(2)

(Total for question = 5 marks)

Q4.

The function f is given by

$$f(x) = 2x^3 - 4$$

(a) Show that $f^{-1}(50) = 3$

(2)

The functions g and h are given by

$$g(x) = x + 2 \text{ and } h(x) = x^2$$

(b) Find the values of x for which

$$hg(x) = 3x^2 + x - 1$$

.....

(4)

(Total for question = 6 marks)

Q5.

For all values of x

$$f(x) = (x + 1)^2 \quad \text{and} \quad g(x) = 2(x - 1)$$

(a) Show that $gf(x) = 2x(x + 2)$

(2)

(b) Find $g^{-1}(7)$

.....

(2)

(Total for question = 4 marks)

Q6.

f and g are functions such that

$$f(x) = \frac{12}{\sqrt{x}} \quad \text{and} \quad g(x) = 3(2x + 1)$$

(a) Find g(5)

.....

(1)

(b) Find gf(9)

.....

(2)

(c) Find $g^{-1}(6)$

.....

(2)

(Total for question = 5 marks)

Q7.
The functions g and h are such that

$$g(x) = \sqrt[3]{2x - 5} \qquad h(x) = \frac{1}{x}$$

(a) Find g(16)

.....

(1)

(b) Find $hg^{-1}(x)$

Give your answer in terms of x in its simplest form.

$hg^{-1}(x)$= .....

(3)

(Total for question = 4 marks)

Q8.
The functions f and g are such that

$$f(x) = 3x - 1 \quad \text{and} \quad g(x) = x^2 + 4$$

(a) Find $f^{-1}(x)$

$f^{-1}(x)$ = ...

(2)

Given that $fg(x) = 2gf(x)$,

(b) show that $15x^2 - 12x - 1 = 0$

(5)

(Total for question = 7 marks)

Q9.

For $x \geq 0$, the functions f and g are such that

$$f(x) = 3x + 4 \qquad g(x) = \frac{\sqrt{x} + 2}{5}$$

(a) Find $g^{-1}(x)$

$g^{-1}(x) =$ .....

(2)

(b) Solve gf $(x) = 3$

$x =$ .....

(3)

(Total for question = 5 marks)

Q11.

$$f(x) = \frac{5x - 3}{4}$$

(a) Find $f^{-1}(x)$

$f^{-1}(x) =$ .....

(2)

For all values of x

$$g(x) = (x - 1)^2 \text{ and } h(x) = 1 - 2x$$

(b) Work out the value of gh(5)

gh(5) = ...

(2)

(Total for question = 4 marks)

LEARNING OUTCOME: To use iterative process to solve equations.

Objectives:

1. To be able to substitute using a calculator.
2. To be able to use iteration to solve equations.

Keywords:
- Sequence
- Iteration

Recurrence Relations/Iterative Sequences

A recurrence relation is a way of defining a sequence that uses the previous term(s) in a sequence to find the next term in the sequence. They can also be defined as iterative methods which means a process that repeats.

The sequence $4, 7, 10, 13, 16$
Has a nth term rule that can be written as: $T_n = 3n + 1$

However, we could write this differently as a recurrence relation, each time to get from one term to the next term all we do is $+3$. Each time we do a calculation we call this an iteration. We write this as:
$U_n = U_{n-1} + 3$, where $U_1 = 4$.

Therefore:
$U_2 = U_{2-1} + 3 = U_1 + 3 = 4 + 3 = 7$
$U_3 = U_2 + 3 = 7 + 3 = 10$
$U_4 =$ _____ $+ 3 =$
$U_5 =$

Using a recurrence relation is easier to use for the first few terms of more complicated sequences but is inefficient to find higher terms in the sequence as they require lots of iterations.

	Example 1- I do…	*Example 2 - We do…*	*Example 3 - You do…*
Steps	A sequence is generated using the following term-to-term formula. x_0 is the first term. $x_0 = 3,$ $x_{n+1} = x_n + 2$ Find the first 3 terms in the sequence.	A sequence is generated using the following term-to-term formula. x_0 is the first term. $x_0 = 3,$ $x_{n+1} = x_n - 5$ Find the first 3 terms in the sequence.	A sequence is generated using the following term-to-term formula. x_0 is the first term. $x_0 = 3,$ $x_{n+1} = x_n + 10$ Find the first 3 terms in the sequence.
Write x_0 into your calculator and click equals.			
Type out the formula and substitute ANS for x_n			
Click equals n times, writing each output down			

INDEPENDENT PRACTICE

1) Find the first 4 terms in each iterative sequence.

a) $x_0 = 3$,
$x_{n+1} = x_n - 7$

$x_1 =$

$x_2 =$

$x_3 =$

$x_4 =$

b) $x_0 = 3$,
$x_{n+1} = x_n + 9$

$x_1 =$

$x_2 =$

$x_3 =$

$x_4 =$

c) $x_0 = 3$,
$x_{n+1} = 2x_n + 9$

$x_1 =$

$x_2 =$

$x_3 =$

$x_4 =$

d) $x_0 = 3$,
$x_{n+1} = (x_n)^2 + 2$

$x_1 =$

$x_2 =$

$x_3 =$

$x_4 =$

APPLICATIONS OF ITERATIVE SEQUENCES

Recurrence relations can also be shown in contextual examples. You have already seen some basic examples of this using compound interest but have never formalised the concept into a recurrence relation.

Example: Money is placed into a bank with an interest rate of 3.5% per annum. Create a recurrence relation for this situation:

Example: £40,000 is placed into a bank account paying interest of 1.1% per month, at the end of every month £200 is deposited into the account. Create a recurrence relation for this situation:

	Example 4- I do…	*Example 5 - We do…*	*Example 6 - You do…*
Steps	The number of slugs in a garden t days from now is p, where $p_0 = 100$, $p_{t+1} = 1.06p_t$. Work out the number of slugs in the garden 3 days from now.	The number of ants in a garden t days from now is p, where $p_0 = 120$, $p_{t+1} = 1.06p_t$. Work out the number of ants in the garden 7 days from now.	The number of flies in a garden t days from now is p, where $p_0 = 1200$, $p_{t+1} = 1.07p_t$. Work out the number of flies in the garden 5 days from now.
Write p_0 into your calculator and click equals.			
Type out the series and substitute ANS for p_n			
Click equals n times (for the amount of terms needed).			

INDEPENDENT PRACTICE

2) Use iteration to find the solution of each problem.

a) The number of slugs in a garden t days from now is p, where $$p_0 = 100$$ $$p_{t+1} = 1.16p_t$$ Work out the number of slugs in the garden 3 days from now.	b) The number of rabbits in a field t days from now is p_t, where $$p_0 = 220$$ $$p_{t+1} = 1.15(p_t - 20)$$ Work out the number of rabbits in the field 5 days from now.	c) The number of people living in a town t years from now is p_t, where $$p_0 = 55000$$ $$p_{t+1} = 1.03(p_t - 800)$$ Work out the number people in the town, 3 years from now.
d) The number of slugs in a garden t days from now is p, where $$p_0 = 200$$ $$p_{t+1} = 1.07p_t$$ Work out the number of slugs in the garden 6 days from now.	e) The number of rabbits in a field t days from now is p_t, where $$p_0 = 520$$ $$p_{t+1} = 1.14(p_t - 10)$$ Work out the number of rabbits in the field 7 days from now.	f) The number of people living in a town t years from now is p_t, where $$p_0 = 40000$$ $$p_{t+1} = 1.08(p_t - 800)$$ Work out the number people in the town 5 years from now.
g) The number of slugs in a garden t days from now is p, where $$p_0 = 100$$ $$p_{t+1} = 1.16p_t$$ Work out the number of slugs in the garden 13 days from now.	h) The number of rabbits in a field t days from now is p_t, where $$p_0 = 220$$ $$p_{t+1} = 1.15(p_t - 20)$$ Work out the number of rabbits in the field 18 days from now.	i) The number of people living in a town t years from now is p_t, where $$p_0 = 55000$$ $$p_{t+1} = 1.03(p_t - 800)$$ Work out the number people in the town 14 years from now.

	Example - I do…	*Example - We do…*
Steps	At time t = 0 hours a tank is full of water. Water leaks from the tank. At the end of every hour there is 2% less water in the tank than at the start of the hour. The volume of water, in litres, in the tank at time t hours is V_t Given that $V_0 = 2000$ $V_{t+1} = kV_t$ write down the value of k.	At the start of year n, the number of animals in a population is P_n At the start of the following year, the number of animals in the population is P_{n+1} where $P_{n+1} = kP_n$ At the start of 2017 the number of animals in the population was 4000 At the start of 2019 the number of animals in the population was 3610 Find the value of the constant k.
Consider the relationship between this period's key numbers and the next period's		
Work out any relevant multipliers (>1 if growing and <1 if depreciating)		
Write as a recurrence relationship if required		

INDEPENDENT PRACTICE

At the start of year n, the number of animals in a population is P_n At the start of the following year, the number of animals in the population is P_{n+1} where $P_{n+1} = kP_n$ At the start of 2017 the number of animals in the population was 8000 At the start of 2020 the number of animals in the population was 4913 Find the value of the constant k.	A hot air balloon is descending. The height of the balloon n minutes after it starts to descend is h_n metres. The height of the balloon (n +1) minutes after it starts to descend, h_{n+1} metres, is given by $h_{n+1} = K \times h_n + 50$ where K is a constant. The balloon starts to descend from a height of 1500 metres at 09 15 At 09 16 the height of the balloon is 1400 metres. Work out the height of the balloon at 09 20

Q1.

The number of slugs in a garden t days from now is p_t where

$p_0 = 100$

$p_{t+1} = 1.06p_t$

Work out the number of slugs in the garden 3 days from now.

.....

(Total for question = 3 marks)

Q2.

A ball is thrown upwards and reaches a maximum height.
The ball then falls and bounces repeatedly.

After the nth bounce, the ball reaches a height of h_n
After the next bounce, the ball reaches a height given by $h_{n+1} = 0.55h_n$

After the 1st bounce, the ball reaches a height of 8 metres.

What height does the ball reach after the 4th bounce?

..... metres

(Total for question = 3 marks)

Q3.

The number of bees in a beehive at the start of year n is P_n.
The number of bees in the beehive at the start of the following year is given by

$$P_{n+1} = 1.05(P_n - 250)$$

At the start of 2015 there were 9500 bees in the beehive.

How many bees will there be in the beehive at the start of 2018?

.....

(Total for question is 3 marks)

Q4.
At the start of year n, the quantity of a radioactive metal is P_n
At the start of the following year, the quantity of the same metal is given by

$$P_{n+1} = 0.87P_n$$

At the start of 2016 there were 30 grams of the metal.

What will be the quantity of the metal at the start of 2019?
Give your answer to the nearest gram.

..... grams

(Total for question = 3 marks)

Q5.
A hot air balloon is descending.
The height of the balloon n minutes after it starts to descend is h_n metres.

The height of the balloon (n +1) minutes after it starts to descend, h_{n+1} metres, is given by

$$h_{n+1} = K \times h_n + 20 \text{ where } K \text{ is a constant.}$$

The balloon starts to descend from a height of 1200 metres at 09 15
At 09 16 the height of the balloon is 1040 metres.

Work out the height of the balloon at 09 18

..... m

(Total for question = 4 marks)

Subtopic 6b – Using Iteration to Estimate Solutions to Equations

Iteration is a way of **estimating solutions** to difficult equations – if you study A-level maths you will learn a lot more about it. For now, you need to know that GCSE questions generally follow a predictable pattern in what you are asked to do. First, you will be asked to 'show there is a solution' (or 'root') in a given interval (i.e. between two numbers). The following examples show how to answer this:

Steps	Show the equation $\sqrt{x+1} = \frac{7}{x}$ Has a solution between 3 and 4. ***Example 7- I do…***	Show the equation $x^3 - 3x - 1 = 0$ Has a root between 1 and 2. ***Example 8 - We do…***	Show the equation $x^3 - 24 = 5x$ Has a solution between 3 and 4. ***Example 9 - You do…***
Rearrange to make equal to zero.			
Evaluate the expression for the two given values – show your working			
There should be a positive and negative value – write *"therefore a root in the interval"*			

INDEPENDENT PRACTICE 3) Use iteration to….

a) Show that $x - \sqrt{x} - 1 = 0$ has a root between 2.6 and 2.7	b) Show that $x^3 + 4x = 1$ has a root between 0 and 1	c) Show that $4x^3 - 2x - 5 = 0$ has a root between 1 and 2
d) Show that $3x - x^3 = -11$ has a root between 2 and 3	e) Show that $x^4 - 5x + 1 = 0$ has a root between 1.5 and 2	f) Show that $x^3 + 5 = 2x$ has a root between -2 and -3

The next steps to estimating solutions are to:

- Rearrange the equation to a given form and then…
- Substitute values into the iteration formula (over and over again)

The second of these is slightly more straightforward so we will practice that before combining with the first part….

Example 10- I do…

The iterative formula
$x_{n+1} = \sqrt{x_n + 7}$ can be used to find an approximate solution for the equation $x^2 - x - 7 = 0$.

Using the iteration formula three times and x_0 = 4 to find an estimate for the solution to 2 decimal places.

Example 11 - We do…

The iterative formula
$x_{n+1} = \sqrt[3]{x_n^2 + 5}$ can be used to find an approximate solution for the equation $x^3 - x^2 - 5 = 0$.

Using the iteration formula four times and x_0 = 2 to find an estimate for the solution to 2 decimal places.

Example 12 - You do…

The iterative formula $x_{n+1} = \frac{3}{1+x_n}$ can be used to find an approximate solution for the equation $x^2 + x - 3 = 0$.

Using the iteration formula three times and x_0 = 1 to find an estimate for the solution to 2 decimal places.

INDEPENDENT PRACTICE 4) Use the iterative formulae to find estimates of solutions for each equation.

a) Starting with $x_0 = 3$, use the iterative formula $x_{n+1} = \frac{7}{x_n^2} + 2$ **three times** to find an estimate for the solution to $x^3 - 2x^2 = 7$

b) Starting with $x_0 = 0$, use the iterative formula $x_{n+1} = \frac{2}{x_n^2+3}$ **three times** to find an estimate for the solution to $x^3 + 3x = 2$

c) Starting with $x_0 = 0$, use the iterative formula $x_{n+1} = \frac{1}{4} + \frac{x_n^2}{4}$ **twice,** to find an estimate for the solution of $x^3 + 4x = 1$

d) Starting with $x_0 = 4.5$, use the iterative formula $x_{n+1} = \sqrt[3]{30 + 10x_n}$ to find an estimate for the solution to $x^3 - 10x = 30$ to 2 decimal places.

e) Starting with $x_0 = 2.5$, use the iterative formula $x_{n+1} = \sqrt[3]{5x_n + 4}$ to find an estimate for the solution to $x^3 - 5x + 4 = 0$ to 2 decimal places.

f) Starting with $x_0 = -2$, use the iterative formula $x_{n+1} = 1 + \frac{11}{x_n - 3}$ to find an estimate for the solution to $x^3 - 4x - 8 = 30$ to 1 decimal place.

ITERATION TO SOLVE EQUATIONS

We now combine our skills of iteration with our prior knowledge of rearranging/changing the subject. This is an unusual step, but you are always given a part (a) and told 'what to aim for' and sometimes you need to try some trial and error before you accurately rearrange as asked. The examples below show this.

Example 13 - I do…

a) Show the equation $x^3 - 1 = 2x$ can be rearranged to give $x = \sqrt[3]{1 + 2x}$

b) Starting with the $x_{0=1.5}$ use the iteration formula $x_{n+1} = \sqrt[3]{1 + 2x_n}$ five times to find an estimate for the solution of $x^3 - 1 = 2x$ that is between 1 and 2.

Round your final answer to 3 decimal places.

Example 14 - We do…

a) Show the equation $x^3 - 24 = 5x$ can be rearranged to give $x = \sqrt[3]{24 + 5x}$

b) Starting with the $x_{0=3.5}$ use the iteration formula $x_{n+1} = \sqrt[3]{24 + 25x_n}$ five times to find an estimate for the solution of $x^3 - 24 = 5x$

Round your final answer to 3 decimal places.

Example 15 - You do…

a) Show that the equation $x^3 - x - 19 = 0$ can be arranged to give $x = \sqrt[3]{x + 19}$

b) Starting with $x_1 = 0$, use the iteration formula $x_{n+1} = \sqrt[3]{x_n + 19}$ four times to find an approximate solution to $x^3 - x - 19 = 0$

Round your answer to 3 decimal places.

c) Substitute your answer to part (b) into the original equation, and comment on the accuracy of the estimated solution.

Example 16- I do…

a) Show that $x^2 + x - 1 = 0$ can be rearranged to make $x = \frac{1}{1+x}$

b) Use the iterative formula $x_{n+1} = \frac{1}{1+x_n}$ three times, starting with $x_1 = 0.5$, to find an approximation to this solution.

c) Explain in words what your answer to part (b) represents.

Example 17 - You do…

a) Show that $x^3 - x^2 - 2 = 0$ can be rearranged to make $x = \sqrt{\frac{2}{x-1}}$

b) Use the iterative formula $x_{n+1} = \sqrt{\frac{2}{x_n - 1}}$ three times, starting with $x_0 = 1.9$, to find an approximation to this solution.

c) Explain in words what your answer to part (b) represents.

INDEPENDENT PRACTICE

4) Use iteration to find the roots in each equation.

i)

(a) Show that the equation $x^3 + 2x = 1$ can be rearranged to give $x = \frac{1}{2} - \frac{x^3}{2}$

(b) Starting with $x_0 = 0$, use the iteration formula $x_{n+1} = \frac{1}{2} - \frac{x_n^3}{2}$ twice to find an estimate for the solution of $x^3 + 2x = 1$ to 2 decimal places.

ii)

(a) Show that the equation

$3x - x^3 = -11$ can be rearranged to give $x = \sqrt[3]{3x + 11}$

(b) Starting with $x_0 = 3$, use the iteration formula $x_{n+1} = \sqrt[3]{3x_n + 11}$ three times to find an estimate for the solution of $3x - x^3 = -11$ to 2 decimal places.

iii)

(a) Show that the equation $x^3 + 5 = 2x$ can be rearranged to give $x = \sqrt[3]{2x - 5}$

(b) Starting with $x_0 = -2.5$, use the iteration formula

$$x_{n+1} = \sqrt[3]{2x_n - 5}$$

Six times to find an estimate for the solution of $x^3 + 5 = 2x$ to 2 decimal places.

iv)

(a) Show that the equation $x^4 + x^3 = 4$ can be rearranged to give $x = \sqrt[4]{4 - x^3}$

(b) Starting with $x_0 = 1.2$, use the iteration formula $x_{n+1} = \sqrt[4]{4 - x_n^3}$ six times to find an estimate for the solution of $x^4 + x^3 = 4$ to 2 decimal places.

v.

a. Show that $x^2 + 4x - 29 = 0$ can be written in the form $x = \sqrt{29 - 4x}$.

b. Use the iteration formula
$x_{n+1} = \sqrt{29 - 4x_n}$ to find x_4 to 3 decimal places. Start with $x_0 = 0$.

c. Explain what your answer to (b) represents

vi.

a. Show that $x - \sqrt{x} - 1 = 0$ has a root between 2.6 and 2.7

b. Using $x_{n+1} = \sqrt{x_n} + 1$ and $x_0 = 2.6$, find x_3 to 3 decimal places.

Q6.

(a) Use the iteration formula $x_{n+1} = \sqrt[3]{10 - 2x_n}$ to find the values of x_1, x_2 and x_3

Start with $x_0 = 2$

x_1 .....

x_2 .....

x_3 .....

(3)

The values of x_1, x_2 and x_3 found in part (a) are estimates of the solution of an equation of the form $x^3 + ax + b = 0$ where a and b are integers.

(b) Find the value of a and the value of b.

a = ...

b = ...

(1)

(Total for question = 4 marks)

Q7.

(a) Show that the equation $x^3 + 2x - 6 = 0$ has a solution between $x = 1$ and $x = 2$

(2)

(b) Show that the equation $x^3 + 2x - 6 = 0$ can be rearranged to give

$$x = \frac{6}{x^2 + 2}$$

(1)

(c) Starting with $x_0 = 1.45$

use the iteration formula $x_{n+1} = \dfrac{6}{x_n^{\,2} + 2}$ twice to find an estimate for the solution of $x^3 + 2x - 6 = 0$

Give your answer correct to 4 decimal places.

.....

(3)

(Total for question = 6 marks)

Q8.

(a) Show that the equation $x^3 + 4x = 1$ has a solution between $x = 0$ and $x = 1$

(2)

(b) Show that the equation $x^3 + 4x = 1$ can be arranged to give

$$x = \frac{1}{4} - \frac{x^3}{4}$$

(1)

(c) Starting with $x^0 = 0$, use the iteration formula $x_{n+1} = \frac{1}{4} - \frac{x_n^{\ 3}}{4}$ twice, to find an estimate for the solution of $x^3 + 4x = 1$

.....

(3)

(Total for question = 6 marks)

Q9.

(a) Show that the equation $x^3 - 3x^2 + 3 = 0$ has a solution between $x = 2$ and $x = 3$

(2)

(b) Show that the equation $x^3 - 3x^2 + 3 = 0$ can be rearranged to give $x = \sqrt[3]{3x^2 - 3}$

(1)

(c) Starting with $x_0 = 2$, use the iteration formula $x_{n+1} = \sqrt[3]{3x_n^{\ 2} - 3}$ to find the value of x_2
Give your answer correct to 3 decimal places.

.....

(3)

(Total for question = 6 marks)

Q10.

(a) Show that the equation $x^4 - x^2 - 5 = 0$ can be written in the form $x = \sqrt[4]{x^2 + 5}$

(1)

(b) Starting with $x_0 = 1.5$

use the iteration formula $x_{n+1} = \sqrt[4]{x_n^2 + 5}$ three times to find an estimate for a solution of $x^4 - x^2 - 5 = 0$

.....

(3)

(Total for question = 4 marks)

Q11.

(a) Show that the equation $3x^2 - x^3 + 3 = 0$ can be rearranged to give

$$x = 3 + \frac{3}{x^2}$$

(2)

(b) Using

$$x_{n+1} = 3 + \frac{3}{x_n^2} \quad \text{with } x_0 = 3.2,$$

find the values of x_1, x_2 and x_3

.....

(3)

(c) Explain what the values of x_1, x_2 and x_3 represent.

.....

.....

(1)

(Total for question is 6 marks)

Q12.

(a) Show that the equation $2x^3 + 4x = 3$ has a solution between 0 and 1

(2)

(b) Show that $2x^3 + 4x = 3$ can be rearranged to give $x = \frac{3}{4} - \frac{x^3}{2}$

(1)

(c) Starting with $x_0 = 0$, use the iteration formula $x_{n+1} = \frac{3}{4} - \frac{x_n^{\ 3}}{2}$ times to find an estimate for the solution to $2x^3 + 4x = 3$

.....

(3)

(Total for question = 6 marks)

Q16.

(a) Show that the equation $x^3 + x = 7$ has a solution between 1 and 2

(2)

(b) Show that the equation $x^3 + x = 7$ can be rearranged to give $x = \sqrt[3]{7 - x}$

(1)

(c) Starting with $x_0 = 2$,

use the iteration formula $x_{n+1} = \sqrt[3]{7 - x_n}$ three times to find an estimate for a solution of $x^3 + x = 7$

.....

(3)

(Total for question = 6 marks)

Subtopic 7 – Algebraic Proof

LEARNING OUTCOME: To understand algebraic proof.

Objectives:

1. To be able to prove statements algebraically.

Keywords:
- Proof

You will have encountered the word "proof" many times in your life. To prove something mathematically can be very challenging. You formally met this concept in a tricky context in in Year 10 with the algebraic approach to proving that **recurring decimals can be written as equivalent fractions**. GCSE exams may ask you to prove other statements are true.

Direct algebraic proof works by manipulating one side of a **mathematical identity** in order to make it exactly equal to the other to the other side. Whilst working through these proofs we will refer to the LHS (left hand side) and RHS (right hand side) of the identity, the first statement we make is 'Consider the LHS'. We then manipulate the LHS until it equals the RHS. This is a helpful introduction to the idea of proof.

	Example 1- I do…	*Example 2 - We do…*	*Example 3 - You do…*
Steps	Prove that $7(x-9)-3(x-4)\equiv 4x-51$	Prove that $(y-4)(y+5)\equiv y^2+y-20$	Prove that $5(x-3)-4(x-2)\equiv x-7$
Consider the side of the identity that is most obvious to manipulate			
Simplify by expanding and collecting like terms.			
Write a statement that you have proven as required – you can write QED to save ink!			

INDEPENDENT PRACTICE

1. Prove the following statements are true:

a. $6(x-7)+5x\equiv 11x-32$	b. $6(x-4)-5(x-2)\equiv x-14$	c. $(x+2)(x-4)-(x+3)(2x-2)\equiv -x^2-6x-2$

ALGEBRAIC PROOF: PRINCIPLES OF NOTATION

To become adept at writing and forming mathematical proofs you need to first understand the basic principles that all mathematicians will use:

- Any integer will be denoted by the variable n.
- Any integer multiplied by 2 becomes an even number, therefore we use $2n$ to represent any even number. If you are asked about "any two even numbers" then use $2n$ and $2m$
- One more or less than any even number is an odd number, therefore $2n \pm 1$ represents any odd number. A 'different' odd number could then be $2m + 1$
- If n is used to denote any integer then a series of consecutive integers would be $n - 1, n, n + 1, n + 2, n + 3, \ldots$
- If $2n$ is used to denote any even number then a series of consecutive even integers would be $2n - 2$, $2n,\ 2n + 2,\ 2n + 2, \ldots$
- If $2n + 1$ is used to denote any odd number then a series of consecutive odd integers would be $2n - 1, 2n + 1,\ 2n + 3, \ldots$

It follows from the above that if you can factorise a "2" out of an algebraic expression, then you have shown that the original expression was even. This can be seen in some of the examples below:

	Example 4 - I do…	*Example 5 - We do…*	*Example 6 - You do…*
Steps	Prove that the sum of an odd number and an odd number is always an even number.	Prove that $5(x - 3) + 3(x + 1)$ is even for all values of n.	Prove that the sum of an odd number and an even number is an odd number.
Write the statements/terms in algebraic form			
It may help to write *"let n be an integer"*			
Write a statement that you have proven as required – you can write QED to save ink!	Prove that $(2n + 1)^2$ is odd for all values of n.	Prove that the difference between the squares of two consecutive even numbers is even.	Prove that $(n + 1)^2 - n^2$ is odd for all values of n.

ALGEBRAIC PROOF: PROOF OF DIVISIBILITY

If a term or an expression is divisible by an integer k, it can be written as kn or $k(\dots)$ where k is factorised out as a factor.

For example, if we are told that n is an integer then to show that $20n - 12$ is divisible by 4 we can factorise out the 4 as follows:

And write our conclusion as follows: _____

INDEPENDENT PRACTICE

1. Prove that the following expressions are divisible by the factors listed in **bold**.

2	$14 - 6n$	**3**	$12n^2 - 6n - 9$	**11**	$121n - 88$
5	$30n - 25$	6	$42 - 36n^2$	**10**	$420 - 50n$
7	$7n^2 - 35$	8	$8 - 16n^2 + 24n$	**2**	$26n^2 - 6 + 18n$

	Example 1- I do…	*Example 2 - We do…*	*Example 3 - You do…*
Steps	Prove that $5(x-3) + 3(x+1)$ is divisible by 4.	Prove that the sum of three consecutive **even** numbers is a multiple of 6.	Prove that the sum of any three consecutive integers is a multiple of 3.
Write out the values algebraically using n.			
Simplify by collecting like terms.			
Factorise.			
Prove with a statement.			

Steps	*Example 4- I do…*
1) Write out the values algebraically using n. 2) Simplify by collecting like terms. 3) Factorise (if it is required) 4) Prove with a statement.	Prove algebraically that the sum of any two consecutive integers is always an odd number
	Example 5 - We do…
	Prove that the difference between the squares of two consecutive odd numbers is a multiple of 8.
	Example 6 - You do…
	n is an integer. Prove algebraically that the sum of n(n + 1) and n + 1 is always a square number.

INDEPENDENT PRACTICE

1) Prove algebraically that the sum of two consecutive odd numbers is divisible by 4

2) I think of two consecutive integers. Prove that the difference of the squares of these integers is equal to the sum of the two integers.

3) Prove $(n + 6)^2 - (n + 2)^2$ is always a multiple of 8

4) Prove $(n + 10)^2 - (n + 5)^2$ is always a multiple of 5

5) Prove that $(3n + 1)^2 - (3n - 1)^2$ is always a multiple of 12, for all positive integer values of n.

6) Prove $(2n + 9)^2 - (2n + 5)^2$ is always a multiple of 4

7) Prove that when two consecutive integers are squared, that the difference is equal to the sum of the two consecutive integers
8) Prove that $(2n + 1)(3n - 2) - (6n - 1)(n - 2)$ is always even
9) Prove the product of three consecutive odd numbers is odd

10) Prove algebraically that $(4n + 1)^2 - (2n - 1)$ is an even number for all positive integer values of n.

11) Prove that the sum of the squares of any three consecutive odd numbers is always 11 more than a multiple of 12.

12) Prove that the difference between two consecutive cubes is one more than a multiple of 6.

13) Prove that there is exactly one sequence of five consecutive positive integers in which the sum of the squares of the first three integers is equal to the sum of the squares of the other two integers.

14) Prove that the product of four consecutive numbers is one less than a square number.

ALGEBRAIC PROOF: DIGIT PROOFS

Digit proofs work by using place value to prove certain conclusions about numbers. The first thing we will do is practise writing numbers algebraically using place value.

Example: $abcd$ is a four-digit number, write down the value of $abcd$ using place value.

Your Turn: $4a1xy$, is a five-digit number, write down the value of $421xy$ using place value

Steps	*Example 4- I do…*
1) Write out the numbers in full place value form 2)Read the question carefully to work out the algebraic manipulation required 3) Factorise (if it is required) 4) Prove with a statement.	Given that $abcd$ is a 4-digit number, prove that the difference between itself and its reverse is a multiple of 11
	Example 5 - We do…
	I think of a two-digit number. I then reverse the digits. Prove that the difference between the two numbers is a multiple of 9
	Example 6 - You do…
	I think of a 3-digit number and then reverse its digits. Prove that the difference between these two numbers is a multiple of 11

Q1.

Given that n can be any integer such that $n > 1$, prove that $n^2 - n$ is never an odd number.

(Total for question = 2 marks)

Q2.

An expression for the nth term of the sequence of triangular numbers is $\frac{n(n+1)}{2}$

Prove that the sum of any two consecutive triangular numbers is a square number.

(Total for question = 3 marks)

Q3.

Prove algebraically that the sum of the squares of any two consecutive even numbers is always a multiple of 4

(Total for question = 3 marks)

Q4.

(a) Prove that

$$(2m + 1)^2 - (2n - 1)^2 = 4(m + n)(m - n + 1)$$

(3)

Sophia says that the result in part (a) shows that the difference of the squares of any two odd numbers must be a multiple of 4

(b) Is Sophia correct?

You must give reasons for your answer.

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(1)

(Total for question = 4 marks)

Q5.

n is an integer.

Prove algebraically that the sum of $\frac{1}{2}n(n+1)$ and $\frac{1}{2}(n+1)(n+2)$ is always a square number.

(Total for question = 2 marks)